
Mitigating factual hallucinations

What we are researching, why the study is designed this way, and where implementation stands

Current position: the local model/runtime and scientific admission path are validated. Prompt-only baseline generation is 46.4% complete. Later steering phases have substantial software support, but no intervention result may be claimed yet.

Reliability means avoiding two opposite failures

Confident fabrication

The model gives a fluent factual answer that is wrong. The user may trust it precisely because it sounds authoritative.

What we call a hallucination

An *incorrect attempted answer*: the response makes a factual commitment that does not match the accepted answer.

Unnecessary refusal

The model says “I do not know” even though it could answer correctly. Hallucinations fall, but usefulness falls with them.

Why error rate alone misleads

A model that refuses every question has zero attempted hallucinations and zero practical utility.

Research goal: answer when the model is likely correct, intervene when knowledge may be recoverable, and abstain only when residual risk remains too high.

We frame the task as selective prediction

Optimization target

Maximize answer coverage subject to a strict bound on incorrect attempted answers, while preserving utility and safety.

$$\max \text{Coverage} \quad \text{subject to} \\ \text{HallucinationRisk} \leq \epsilon$$

Example: zero errors in 1,000 attempted answers implies an approximate one-sided 95% upper risk bound of $3/1000 = 0.3\%$, not universal zero risk.

Why this formulation

- It makes refusal cost visible.
- It supports fair comparisons at matched coverage.
- It separates “fewer errors” from “more knowledge.”
- It allows a statistical upper bound when zero errors are observed.

Four questions determine whether steering is genuinely useful

RQ1 — Adaptive vs. static

Can the query choose its vector, layer, and strength better than one fixed direction?

RQ2 — Knowledge vs. abstention

Do incorrect answers become correct, or merely become refusals?

RQ3 — Sparse and protected edits

Can factuality features be changed without harming language, safety, or general capability?

RQ4 — Prompt interaction

Does internal steering add value beyond inexpensive textual instructions?

Why all four are needed: a lower raw hallucination count can otherwise be explained by stronger perturbations, more refusals, benchmark-specific tuning, or a prompt that already produces the same effect.

Steering changes behavior inside inference

What it is

We read an internal activation, add a learned direction at a chosen transformer site, and continue generation without retraining all model weights.

Methods being compared

M0 no steering; M1 dense centroid; M2 published baselines; M3 adaptive controller; M4 sparse features; M5 protected subspace; M6 composite policy.

Why go beyond one global vector

A dense direction can mix factuality with uncertainty, refusal style, language, and safety. Applying it to every query may therefore fix some errors while creating new regressions.

Planned improvement

Estimate risk from the current query, intervene only in the recoverable region, and constrain the edit to features that preserve protected behavior.

Core causal test: compare every intervention with the same checkpoint, questions, prompts, decoding, and grader; change only the internal policy.

Outcome transitions reveal what an improvement means

Response labels

C — correct
P — partially correct
I — incorrect attempted
A — abstention
U — unscorable

Desired transitions

$I \rightarrow C$: knowledge recovery
 $I \rightarrow A$: safer abstention
 $C \rightarrow C$: correct preservation

Failure transitions

$C \rightarrow A$: strict over-refusal
 $C \rightarrow I$: factual regression
correct $\rightarrow$ wrong language or unsafe behavior

Why forced-answer and likelihood tests matter: if gains disappear when abstention is discouraged and gold-answer likelihood does not rise, steering probably changed refusal behavior rather than knowledge expression.

Three benchmarks serve different scientific roles

Benchmark	Role	Why it is included
TriviaQA	5,000-question controller split	Large development source for probes, vectors, routers, and in-distribution testing; evidence documents are discarded to keep the task closed-book.
SimpleQA Verified	1,000 untouched questions	External short-answer test with an explicit correct/incorrect/not-attempted rubric; useful for detecting refusal-driven gains.
AA-Omniscience Public-600	600 untouched questions	Directly tests calibrated knowledge boundaries and includes an official abstention-aware setup. Public size limits domain-level claims.

Why this split matters: methods are learned on TriviaQA and transferred unchanged to the two evaluation sets, reducing target-benchmark overfitting.

Resource limits changed the execution platform

What happened

- 1 Colab Free offered a T4 or v5e TPU, but its RAM/session limits could not support the complete activation workflow reliably.
- 2 The Colab notebook and import bundle were retired before scientific E1 results.
- 3 The study moved to one native Apple-silicon checkpoint on the available MacBook Pro.

Why this was the defensible choice

- Stable access to internal hooks is more important than nominal accelerator speed.
- Local runs can resume without session eviction.
- One pinned representation prevents mixing incompatible activations across runtimes.
- The causal questions remain testable within one model.

Limitation: conclusions apply only to this checkpoint/runtime; cross-model replication remains future work.

One-bit Bonsai makes local intervention feasible

Pinned system

`prism-ml/Bonsai-27B-mlx-1bit`

exact revision `ef22f239...b96`

Prism MLX fork `10b5fe43...4b7`

Apple M1 Pro, 16 GiB unified memory

64 blocks, hidden size 5,120; pinned snapshot
5.16 GB; custom grouped binary
representation averaging 1.125
language-weight bits.

Why this model

- It fits alongside inference state on the available machine.
- Native MLX preserves the released one-bit weights.
- The fork exposes post-attention residual, MLP output, and block output intervention sites.
- Cached continuation can be captured and steered, which is essential for the planned experiments.

Human review closes a gap automation cannot settle

What we did

Exact, fuzzy, and semantic matching created a blinded queue of the 200 highest-risk contamination candidates. A human reviewed the supplied CSV without model-condition information.

Result

200 rows reviewed; 197 distinct candidate pairs; 3 genuine overlaps were excluded; none affected the 500-question E0 cohort.

Why a person was needed

Similar wording, shared entities, and paraphrases can look like duplicates to an embedding model without asking the same factual question. Conversely, true leakage can be phrased differently.

Why it gates E1

Vectors and controllers must not learn from evaluation questions. The signed review receipt proves the development/evaluation boundary before expensive computation starts.

External graders are used where exact matching is insufficient

What we implemented

Local deterministic generation is followed by frozen OpenRouter grading for semantic benchmarks: GPT-4.1 with the released SimpleQA rubric and stable Gemini 2.5 Flash with the AA rubric.

The adapter records the prompt, model identity, transport, retries, parser result, and terminal failures.

The retired Gemini preview endpoint was replaced before E1 grading; the amendment preserves the rubric but explicitly limits checkpoint comparability.

Why this is necessary

Correct answers often use aliases or explanatory wording that string matching cannot judge. AA also needs a partial-credit distinction.

Why it is frozen and fail-closed

A changing grader can change the experiment. Atomic labels, bounded retries, and “unscorable” on failure prevent silent defaults from biasing results.

Content-addressed gates keep runs auditable

What was built

Every phase has frozen inputs, manifests, SHA-256 identities, append-only records, resumable chain heads, prerequisite receipts, and replayable finalizers.

Model, tokenizer, prompt template, decoding, hardware, runtime, split, grader, and code identities are bound to each run.

Why this is unusually strict

- Multi-day local runs must resume without duplicating or silently changing rows.
- Later phases must not consume unverified upstream outputs.
- Hyperparameters must be selected before untouched tests are opened.
- A result should be independently replayable from immutable evidence.

Interpretation rule: passing tests shows that software enforces the protocol; only a completed scientific ledger supports an experimental claim.

E0 tests whether local experimentation is viable

What E0 tested

- exact model and snapshot identity
- tokenizer and chat-template identity
- deterministic decoding across repeated questions
- throughput, latency, and peak unified memory
- all required activation capture/intervention sites
- scientific admission after contamination review

Design choice: 500 model-independent reserved questions, each generated twice with the exact E1 decoding path.

Why E0 came first

The rest of the plan depends on stable activations. If the checkpoint, template, cached path, or hook changes, learned vectors no longer refer to the same internal system.

Measuring memory before activation collection also prevents designing E2–E8 around an impossible storage or compute budget.

E0 passes every runtime gate

500

questions, each repeated twice

0

determinism mismatches

13.84

mean generated tokens/s

4.86 GB

measured peak memory

Passed

- ✓ checkpoint
- ✓ chat template
- ✓ deterministic decode
- ✓ MLX runtime

Meaning: the 27B one-bit model leaves enough memory headroom and produces repeatable outputs on the exact local path required by later phases.

Not an accuracy result: E0's strict whole-response alias match was only a runtime diagnostic; its 1/500 match must not be reported as benchmark accuracy.

E1 establishes the prompt-only baseline

Frozen experiment

One model $\times$ three benchmarks $\times$ three prompts:

P0 neutral; P1 direct answer; P2 calibrated abstention.

19,800 deterministic local generations, maximum 48 new tokens, seed 17; 4,800 semantic grades scheduled after generation.

Why prompt-only comes first

A calibrated instruction may reproduce part of steering's benefit at almost no engineering cost. Without E1, later gains cannot be attributed to internal intervention.

E1 also supplies the C/P/I/A labels used to learn probes, construct correct-versus-incorrect activation pairs, and power later paired analyses.

E1 generation is 46.4% complete

9,184 of 19,800 generations are provenance-bound

46.4%

10,616 remaining

- append-only rows with a resumable chain head
- condition-major order across all prompt/benchmark cells
- exact checkpoint, split, prompt, runtime, and protocol hashes
- grading deliberately waits on frozen local outputs

Live checkpoint: `artifacts/checkpoints/E1-generation.json`, read 17 July 2026.

Evidence boundary

Can claim: computation is active, deterministic, resumable, and identity-bound.

Cannot claim: accuracy, hallucination risk, abstention, prompt effects, or steering gains.

E2 locates internal factual-outcome separation

What is implemented

Activation capture at validated sites, compact sharding, calibrated risk probes, layer screening, immutable selection artifacts, and replayable finalization.

Why E2 exists

Steering should target a location where correct and incorrect outcomes are measurably separable. Choosing a layer first and testing it later would invite cherry-picking.

Status: software and replay paths are validated; scientific capture remains blocked until E1 supplies verified outcome labels.

Static and established baselines come first

E3 — static steering

Dense correct-minus-incorrect vectors, layer–alpha–token-scope search, random controls, and label-permutation controls.

Why: establish the simplest causal intervention and test whether changes exceed arbitrary perturbation.

Status: independently accepted at the software/replay level; experimental execution remains blocked by E1.

E4 — established methods

Capability reports and execution receipts for CAA, ITI, ACT, SADI, and TruthX, followed by screening and promotion gates.

Why: compare against prior approaches before claiming that the new adaptive controller adds value.

E5 and E6 contain the core causal tests

E5 — adaptive steering: no-go

A 972-arm matched-budget grid and signed execution receipts exist. The remaining blockers are scientifically material: correct M1-R/M1-P identity lookup, split provenance tied to verified E2/E3 artifacts, exact alpha binding, and direct finalizer tests.

Why it matters: otherwise an adaptive win could come from the wrong baseline, leaked split, or unequal intervention budget.

E6 — knowledge vs. refusal: partial

Teacher-forced gold-answer and abstention likelihoods, forced-answer ranks, signed row bindings, and bundle replay exist. The registered knowledge-recovery gate and terminal finalizer remain incomplete.

Why it matters: E6 determines whether E5 changes factual answer selection or only teaches the model to decline.

E7–E10 build the final system

Phase	Planned test	Why it is needed
E7	Coordinate and sparse-autoencoder steering	Test whether a smaller feature set preserves the factuality effect with fewer collateral changes.
E8	Protected subspaces for refusal, safety, and language	Prevent a factuality direction from unintentionally changing protected behavior.
E9	Full prompt $\times$ method factorial	Measure prompt-only, steering-only, combined, and interaction effects under paraphrases.
E10	Frozen composite policy and untouched confirmation	Combine only independently selected parts, run once, and estimate risk–coverage with utility/safety constraints.

Current state: low-level methods, contracts, metrics, and gates are scaffolded; end-to-end orchestration and scientific runs remain.

Viability is established; efficacy remains open

Established

The exact 27B one-bit model runs deterministically on the M1 Pro, fits memory, exposes required intervention sites, and passed contamination admission.

In progress

Prompt-only baseline generation is 46.4% complete. Grading and baseline metrics have not yet been finalized.

Implemented or scaffolded

Later-phase code encodes the planned tests, provenance, and replay rules. It is not evidence that steering improves factuality.

Honest current conclusion: we have de-risked the experimental platform and frozen a rigorous baseline. The central scientific hypotheses remain open.

Next: complete the evidence chain

1. Complete and grade E1

Finish 10,616 local generations, execute 4,800 grades, audit disagreements, derive baseline C/P/I/A distributions, and freeze the E1 receipt.

2. Unlock E2–E4 in order

Capture only frozen activations, calibrate probes, select static controls, and compare established baselines without touching confirmatory sets.

3. Repair E5 and close E6

Fix identity/split/alpha bindings and finalizer tests; then prove knowledge recovery with transitions, likelihoods, and forced answering.

4. Promote claims only after replay

Require prerequisite ledgers, registered gates, immutable artifacts, statistical analysis, side-effect checks, and independent validation.

Final decision rule: success means lower hallucination risk at meaningful matched coverage, with correct answers preserved and no material safety, language, or utility regression.